

Introduction to Computer Programming, Grade 11

College Preparation

ICS3C

This course introduces students to computer programming concepts and practices. Students will write and test computer programs, using various problem-solving strategies. They will learn the fundamentals of program design and apply a software development life-cycle model to a software development project. Students will also learn about computer environments and systems, and explore environmental issues related to computers, safe computing practices, emerging technologies, and post-secondary opportunities in computer-related fields.

Prerequisite: None

A. PROGRAMMING CONCEPTS AND SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate the ability to use different data types in expressions in simple computer programs;
- A2.** demonstrate the ability to use control structures and simple algorithms in computer programs;
- A3.** use proper code maintenance techniques and conventions when creating computer programs.

SPECIFIC EXPECTATIONS

A1. Data Types and Expressions

By the end of this course, students will:

- A1.1** use constants and variables, including integers, floating points, strings, and Boolean values, correctly in computer programs;
- A1.2** demonstrate the ability to manipulate string data in a computer program (*e.g., swap two characters, capitalize first letter, extract a portion of an address, count the occurrences of a word or letter*);
- A1.3** use assignment statements correctly with both arithmetic and string expressions in computer programs (*e.g., numStudents = 4 + 2, name = "Devi"*);
- A1.4** use Boolean operators (*e.g., AND, OR, NOT*), comparison operators (*i.e., equal to, not equal to, greater than, less than, greater than or equal to, less than or equal to*), arithmetic operators (*e.g., addition, subtraction, multiplication, division, exponentiation, parentheses*), and order of operations correctly.

A2. Control Structures and Simple Algorithms

By the end of this course, students will:

- A2.1** write programs that incorporate user input, processing, and screen output;
- A2.2** use sequence, selection, and repetition control structures to create programming solutions;
- A2.3** demonstrate the ability to write algorithms with nested structures.

A3. Code Maintenance

By the end of this course, students will:

- A3.1** explain the difference between syntax, logic, and run-time errors in computer programs;
- A3.2** demonstrate the ability to correct syntax, logic, and run-time errors in computer programs;
- A3.3** use workplace and professional conventions (*e.g., naming, indenting, commenting*) correctly to write programs and internal documentation;
- A3.4** demonstrate the ability to interpret error messages displayed by programming tools (*e.g., compiler, debugging tool*), at different times during the software development process (*e.g., writing, compilation, testing*);
- A3.5** demonstrate the ability to validate a program using test cases.

B. SOFTWARE DEVELOPMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use a variety of problem-solving strategies to solve different types of problems;
- B2.** design software solutions to meet a variety of challenges, using a set of standards;
- B3.** design simple algorithms according to specifications;
- B4.** apply a software development life-cycle model to a software development project.

SPECIFIC EXPECTATIONS

B1. Problem-solving Strategies

By the end of this course, students will:

- B1.1** use various problem-solving strategies (*e.g., divide and conquer, working backwards, process analysis, examples, extreme cases, tables and charts, trial and error*) to solve programming problems;
- B1.2** use the input-process-output model to solve programming problems.

B2. Designing Software Solutions

By the end of this course, students will:

- B2.1** design a simple program from a program template or skeleton (*e.g., teacher-supplied skeleton, Help facility code snippet*);
- B2.2** use appropriate vocabulary and mode of expression (*i.e., written, oral, diagrammatic*) to describe alternative program designs and to explain the structure of a program;
- B2.3** write subprograms (*e.g., functions, procedures*) that perform one well-defined task and use parameter passing and appropriate variable scope (*e.g., local, global*);
- B2.4** use industry-standard programming tools (*e.g., structure chart, flow chart, UML [Unified Modeling Language], data flow diagram, pseudo-code*) to represent the structure and components of a computer program;
- B2.5** design user-friendly software interfaces (*e.g., prompts, messages, screens, forms*).

B3. Designing Simple Algorithms

By the end of this course, students will:

- B3.1** use simple algorithms (*e.g., validate entered data, count, accumulate, use random numbers*) to design a program according to specifications;
- B3.2** solve problems (*e.g., calculation of gross pay; fuel consumption on a car trip; average of students' marks; temperature at a given altitude, using the environmental lapse rate*) by applying mathematical equations or formulas in an algorithm;
- B3.3** design algorithms to detect, intercept, and handle run-time errors (*e.g., division by zero, roots of negatives*).

B4. The Software Development Life Cycle

By the end of this course, students will:

- B4.1** describe the phases (*i.e., problem definition, analysis, design, writing code, testing, implementation, maintenance*), milestones (*e.g., date of completion of program specification*), and products (*e.g., specification, flow chart, program, documentation, bug reports*) of a software development life cycle;
- B4.2** use a variety of techniques (*e.g., dialogue, questionnaires, surveys, research*) to clarify program specifications;
- B4.3** use project management tools (*e.g., Gantt chart, critical path diagram, PERT chart*) to show tasks and milestones in a teacher-led project;

- B4.4** use a test plan to test programs (i.e., identify test scenarios, identify suitable input data, calculate expected outcomes, record actual outcomes, and conclude 'pass' or 'fail') by comparing expected to actual outcomes;
- B4.5** use a variety of methods to debug programs (*e.g., manual code tracing, extra code to output the state of variables*);
- B4.6** communicate information about the status of a project (*e.g., milestones, work completed, work outstanding*) effectively in writing throughout the project.

C. COMPUTER ENVIRONMENTS AND SYSTEMS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of the functions of different types of computer components;
- C2.** use appropriate file maintenance practices to organize and safeguard data;
- C3.** use a software development environment to write and run computer programs.

SPECIFIC EXPECTATIONS

C1. Computer Components

By the end of this course, students will:

- C1.1** describe the functions and features of the internal components of a computer (*e.g., CPU, RAM, ROM, cache, hard drive, motherboard, power supply, video card, sound card*);
- C1.2** use correct terminology to describe computer features and specifications (*e.g., processor type, bus speed, storage capacity, amount of memory*);
- C1.3** describe the functions and features of common computer peripheral devices (*e.g., printer, monitor, scanner, keyboard, mouse, speakers, USB flash drive*);
- C1.4** compare and contrast common ISP services (*e.g., DSL, cable, dial-up, regional Wi-Fi*) and home networking hardware (*e.g., NICs, routers, hardware used for wired and wireless connections*).

C2. File Maintenance

By the end of this course, students will:

- C2.1** use an operating system to logically organize computer files for easy retrieval, backup, and recovery;
- C2.2** use standard backup procedures to back up user files.

C3. The Software Development Environment

By the end of this course, students will:

- C3.1** describe the functions and features of a software development environment and use it to write and run a computer program;
- C3.2** describe the differences between applications, programming languages, and operating systems;
- C3.3** use Help documentation as a guide to designing and writing programs.

D. COMPUTERS AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** describe computer use policies that promote environmental stewardship and sustainability;
- D2.** describe and apply procedures for safe computing to safeguard computer users and their data;
- D3.** explain key aspects of the impact that emerging technologies have on society;
- D4.** describe postsecondary education and career prospects related to computer studies.

SPECIFIC EXPECTATIONS

D1. Environmental Stewardship and Sustainability

By the end of this course, students will:

- D1.1** describe negative effects of computer use on the environment (*e.g., creation of waste, unnecessary printing of emails, heavy power consumption*) and on human health (*e.g., exposure to radiation, musculoskeletal disorders, eye strain, various health consequences of reduced activity levels*);
- D1.2** identify measures that help reduce the impact of computers on the environment (*e.g., lab regulations, school policies, corporate policies, provincial policies, paperless workplaces, computer recycling and reuse*) and on human health (*e.g., ergonomic standards*);
- D1.3** describe ways in which computers are or could be used to reduce resource use and to support environmental protection measures (*e.g., computer modelling to reduce use of physical resources; interpretation of large amounts of environmental data; management of natural resources; programmable temperature control to reduce energy consumption*);
- D1.4** identify government agencies and community partners that provide environmental stewardship opportunities (*e.g., local community recycling centres, private companies that refurbish computers, printer cartridge recycling programs*).

D2. Safe Computing

By the end of this course, students will:

- D2.1** explain the need for an acceptable-use policy for using computers at school and at work;

- D2.2** describe and use appropriate strategies to avoid potential health and safety problems associated with computer use (*e.g., musculoskeletal disorders, eye strain*);

- D2.3** describe procedures to safeguard data and programs from malware (*e.g., viruses, spyware, adware*).

D3. Emerging Technologies

By the end of this course, students will:

- D3.1** explain how emerging technologies can affect personal rights and privacy (*e.g. video surveillance, cyberbullying, identity theft*);
- D3.2** describe some emerging technologies and their implications for, and potential uses by, various members of society;
- D3.3** describe some of the solutions to complex problems affecting society that have been or are being developed through the use of advanced computer programming and emerging technologies (*e.g., monitoring and regulating electrical supply and demand; using facial recognition programs to verify the identity of persons entering a country; analysing criminal activity by overlaying crime data on satellite imagery; analysing large-scale meteorological data to predict catastrophic storms*).

D4. Postsecondary Opportunities

By the end of this course, students will:

- D4.1** research and describe trends in careers that require computer skills, using local and national sources (*e.g., local newspaper, national newspaper, career websites*);
- D4.2** identify opportunities for experiential learning (*e.g., co-op programs, job shadowing, career fairs*) related to computer science;
- D4.3** research and report on postsecondary educational programs leading to careers in the field of information systems and computer science (*e.g., institutions offering relevant programs, industry certifications, courses of study, entrance requirements, length of programs, costs*);
- D4.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in computer-related fields (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D4.5** describe the Essential Skills and work habits that are important for success in computer studies, as identified in the Ontario Skills Passport.